

INDIA.RUNS

Build what next India runs on

Team Name : TalentRank AI

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PROBLEM STATEMENT: Traditional candidate matching systems rely mainly on keyword searches, which can miss highly relevant candidates. The goal of this project is to develop an AI-powered ranking system that analyzes multiple candidate signals and generates a reliable shortlist for recruiters.

Solution Overview

- **What is your proposed solution?**

Our proposed solution is an AI-powered candidate ranking system that evaluates candidates using multiple signals such as experience, job title relevance, skills, career history, recruiter engagement signals, notice period, relocation preference, and location preference. The system assigns a weighted score to each candidate and generates a ranked shortlist of the most suitable candidates for the job role.

- **What differentiates your approach from traditional candidate matching systems?**

Traditional candidate matching systems primarily rely on keyword matching, which often misses strong candidates who may not use exact keywords in their profiles. Our approach evaluates the overall candidate profile, including career progression, technical expertise, hiring readiness, and behavioral signals. By combining multiple candidate attributes into a unified scoring model, the system provides more accurate and recruiter-friendly rankings.

JD Understanding & Candidate Evaluation

- What are the key requirements extracted from the JD?

The Job Description emphasizes candidates with relevant engineering or AI/ML backgrounds, strong technical expertise, industry experience, and readiness to join. Key requirements include:

- Relevant work experience (5–9 years preferred)
- Engineering, AI, ML, Data Science, or Data Engineering roles
- Strong technical and domain skills
- Proven project and career history
- Active job-seeking signals
- Location and relocation flexibility
- Short notice period for faster hiring

- Which candidate signals are most important for determining relevance? / How does your solution evaluate candidate fit beyond keyword matching?

The system evaluates experience, job title, skills, career history, recruiter activity, and profile completeness. Multiple signals are combined to identify the best-fit candidates beyond simple keyword matching.

Ranking Methodology

- **How does your system retrieve, score, and rank candidates?**

The system processes candidate profiles, extracts relevant features, assigns weighted scores, and ranks candidates based on their total score.

- **What models, algorithms, or heuristics are used?**

A rule-based scoring algorithm is used, combining experience, skills, career history, recruiter signals, and hiring readiness factors.

- **How are multiple candidate signals combined into a final ranking?**

Each candidate signal is assigned a weight, and the combined score determines the final ranking order of candidates.

Explainability & Data Validation

- How are ranking decisions explained?

Each candidate receives a score along with reasons showing which factors contributed to their ranking.

- How do you prevent hallucinations or unsupported justifications?

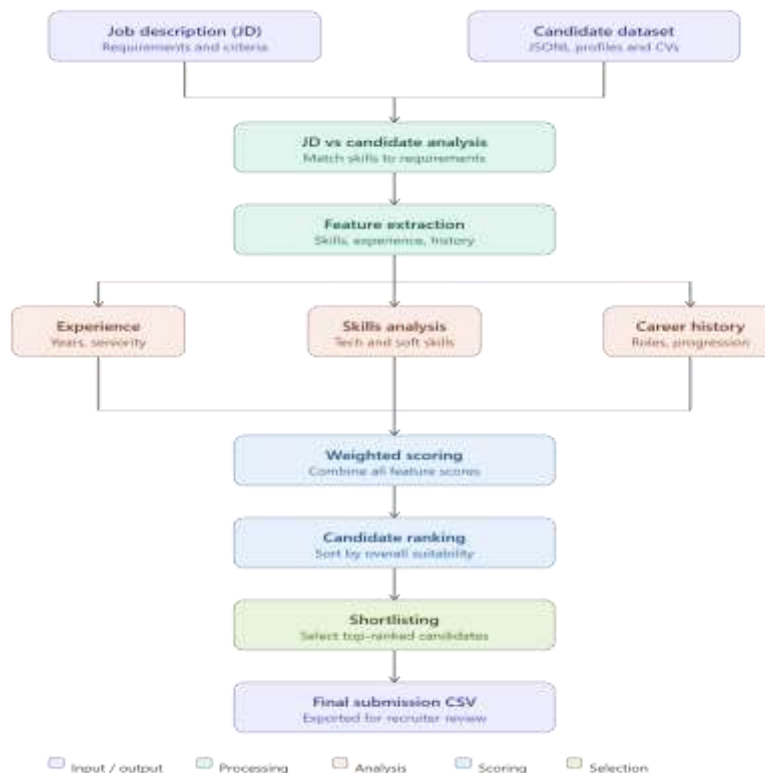
The system uses only available candidate data and predefined scoring rules, ensuring every ranking decision is evidence-based.

- How does your solution handle inconsistent, low-quality, or suspicious profiles?

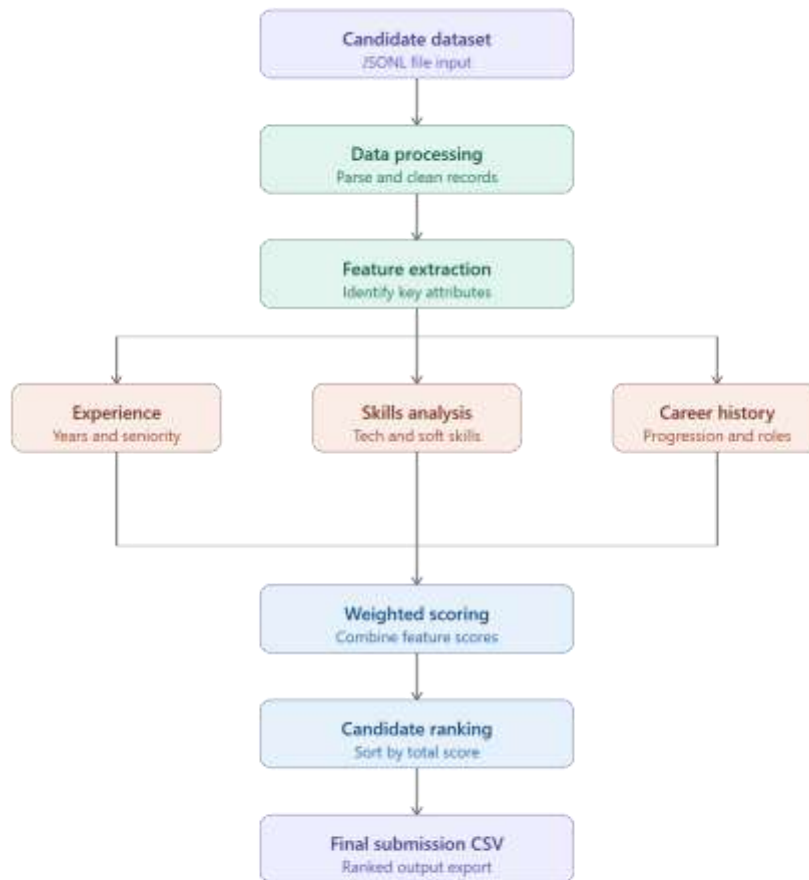
Profiles with missing, incomplete, or weak information receive lower scores, reducing their impact on the final ranking.

End-to-End Workflow

- What is the complete workflow from JD input to ranked candidate output?



System Architecture



Results & Performance

- **What results or insights demonstrate ranking quality?**

The ranking system successfully processed 100,000 candidate profiles and generated a ranked shortlist based on multiple relevance signals. The scoring model effectively prioritized candidates with relevant experience, technical expertise, and strong career progression, helping reduce the chances of overlooking high-quality talent.

By combining multiple candidate signals instead of relying solely on keyword matching, the system produced more accurate and recruiter-friendly rankings. The generated shortlist provides clear and explainable recommendations, making the hiring process faster, more efficient, and easier for recruiters to trust.

- **How does your solution meet the challenge's runtime and compute constraints?**

The solution uses a lightweight rule-based scoring approach implemented in Python, allowing large candidate datasets to be processed efficiently without requiring expensive AI model inference. Feature extraction and score calculation are performed in a single pass through the dataset, ensuring low memory usage and fast execution.

The system successfully processed 100,000 candidate profiles within the challenge constraints while generating accurate and explainable rankings. This approach provides a good balance between ranking quality, runtime performance, and computational efficiency.

Technologies Used

- What technologies, frameworks, and tools were used and why were they selected for this solution?

Category	Technology
Programming Language	Python
Data Processing	JSON / JSONL
Output Generation	CSV
Development Environment	VS Code
Version Control	GitHub

Submission Assets

GitHub Repository Link

Project Presentation (PDF)

final_submission.xlsx

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THANK YOU